Andrew Lee

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Fully Authorized to Work in the U.S. No visa sponsorship needed.

Education

Harvard University — Cambridge, MA
Master of Science in Data Science — *Aug. 2025 – Dec. 2026 (Expected)*

William and Mary — Williamsburg, VA
Bachelor of Science in Mathematics, Honors: Phi Beta Kappa (Top 5%) — *Aug. 2019 – May 2025*

Experience

Data Science Intern — Expected Start: June 2025
E. & J. Gallo Winery — *Modesto, CA*

- Analyze data to transform into useful information to help make business decisions
- Model structured and unstructured data for system integration
- Data mining to identify patterns for predictive insights
- Organizing data using visualization

Product Management Intern — May 2022 – Aug. 2022
Council on International Educational Exchange (CIEE) — *Seoul, South Korea*

- Designed and launched 3 study abroad programs using historical enrollment data and alumni feedback to tailor themes, resulting in a 20% increase in conversion from initial program inquiry to completed applications
- Managed program logistics for 400+ participants by maintaining student records in Excel, resolving data inconsistencies across flights, housing, and itineraries with 30% reduction in coordination issues
- Ran A/B testing on post-event survey data (n=350) to evaluate cost-efficiency of guest speaker formats to show no significant satisfaction difference ($p > 0.05$), influencing a budget pivot and saving ₩2M

Projects

Counting Cranes Using Deep Learning | *PyTorch, OpenCV, MLOps* | [Code]

- Collaborated with U.S. Fish and Wildlife Service to develop object detection models for counting migratory birds, training YOLO-family models on 47,000+ aerial images and achieving a 5% test error
- Deployed an end-to-end MLOps workflow by containerizing a user-friendly application with Docker and deploying on an AWS EC2 instance with a FastAPI backend and Next.js frontend
- Reduced operational costs by an estimated $20,000+ by enabling weeks-long surveys to be completed in hours
- Presented research findings to the CEO of Conservation International and at multiple research seminars

Spatiotemporal Analysis of Wildfire Response | *R, Data Mining* | [Code] [Paper]

- Designed a Bayesian spatiotemporal distributed lag model to forecast public response to wildfire smoke (PM2.5) using Google Trends data of 10 public health-related keywords
- Leveraged Selenium library to automate data collection and conducted exploratory data analysis (EDA) in R.
- Revealed increased sensitivity to smoke PM2.5 among individuals experiencing poverty or lacking health insurance with a 1 SD PM2.5 increase linked to 22% and 18% search surge for "Air Pollution" and "Air purifier"

Machine Learning in Hypersonics | *Scikit-Learn, TensorFlow* | [Code]

- Built and cross-validated supervised learning models (Linear Regression, Random Forest, SVM) to predict Shock-Train Leading Edge (STLE) in hypersonic vehicles using high-dimensional sparse pressure sensor data
- Improved R-squared of 0.64 from a previous study to 0.89 by engineering time-series features to reduce sensor noise
- Trained a feed-forward neural network using limited data (7 of 37 runs) and achieved successful generalization to 900K+ test observations

Visualizing World Happiness | *Tableau, Matplotlib, seaborn, Plotly* | [DashBoard]

- Utilized Python and Tableau to build and deploy an interactive dashboard to visualize global happiness trends

Technical Skills

Python (Pandas, NumPy), SQL, R, PyTorch, AWS (EC2, S3, ECR, Lambda), Git, Docker, DVC, Linux/Bash